

Abstract Algebra

SF1678

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1. Sets

Definition 1.1 (Tuple)

A tuple (or sometimes *list*) is a finite, ordered collection of elements, possibly with repetitions.

An n -tuple is a tuple of length n .

1.0.1. Morphisms

Definition 1.0.2 (Map)

Subsets of the Cartesian product $A \times B$ are called *relations*. A *map* is a *well-defined* relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \rightarrow B \quad \text{or} \quad A \xrightarrow{f} B$$

and is

- surjective* or *onto* if $f(A) = B$, which is equivalent to being *right invertible* with $f \circ f^{-1} = \text{id}_B$.
- injective* or *one-to-one* if $a_1 \neq a_2 \implies f(a_1) \neq f(a_2)$, which is equivalent to being *left invertible* with $f^{-1} \circ f = \text{id}_A$.
- bijective* or a *one-to-one correspondence* if both above apply, which is equivalent to being *invertible*.

Definition 1.0.3 (Endomorphism)

Let X be some mathematical object. A map $\varphi : X \rightarrow X$ is an *endomorphism*. Should φ simultaneously be bijective it is moreover an *automorphism* and, if X is a set, a *permutation*. The most trivial automorphism is the *the identity map* id_X with

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Definition 1.0.4 (Homomorphism)

A *homomorphism* is a map between algebraic structures that preserves the structure.

For example, a group homomorphism $f : G \rightarrow H$ satisfies

$$f(a * b) = f(a) * f(b)$$

for all $a, b \in G$.

Definition 1.0.5 (Isomorphism)

A map $T : V \rightarrow W$ is an *isomorphism* if it is

- (i) bijective and linear or, equivalently,
- (ii) invertible.

If such a map exists, the spaces are *isomorphic*, written $V \cong W$.

Then the spaces differ only in their respective choice of basis.

If $V = W$, then T is an *automorphism*.

Definition 1.0.6 (Kernel)

Let $T : V \rightarrow W$ be a linear map. The *kernel* of T is

$$\ker(T) = \{v \in V \mid T(v) = 0\}.$$

The kernel is a subspace of V .

1.0.2. Equivalence

Definition 1.0.7 (Equivalence Relation)

An *equivalence relation* on a set A is a relation $R \subseteq A \times A$ with the following properties:

- (i) **reflexive** $(a, a) \in R \quad \forall a \in A$
- (ii) **symmetric** $(a, b) \in R \implies (b, a) \in R$
- (iii) **transitive** $(a, b), (b, c) \in R \implies (a, c) \in R$

If $(x, y) \in R$ we write $x \sim y$.

Definition 1.0.8 (Partition)

A *partition* $\mathcal{P}$ of a set A is a collection of non-empty, disjoint subsets of A whose union is A .

Definition 1.0.9 (Equivalence Class)

Let R be an equivalence relation on a set A . The *equivalence class* of $a \in A$ is

$$[a] = \{b \in A \mid b \sim a\}.$$

The set of all equivalence classes is written

$$A \parallel R$$

and is called the *quotient set*.

The equivalence classes form a partition of A .

Definition 1.0.10 (Quotient Map)

Let R be an equivalence relation on A . The map $\pi : A \rightarrow A \parallel R$ defined by

$$\pi(a) = [a]$$

is called the *quotient map*.

It is always surjective.

1.0.3. Transpositions

Definition 1.0.11 (Transposition)

A transposition of the elements $i, j \in \{1, \dots, n\}$ is the permutation τ_{ij} with the cyclic notation (ij) .

Theorem 1.0.1

A permutation $\sigma \in S_n$ can be written as a composition of transpositions.

Definition 1.0.12 (Sign of a Permutation)

Let k be the number of transpositions in any decomposition of a permutation $\sigma \in S_n$. The *sign* of σ is $(-1)^k$. Hence,

- if k is *even* then $\text{sgn}(\sigma) = +1$ and σ is said to be even.
- if k is *odd* then $\text{sgn}(\sigma) = -1$ and σ is said to be odd.

Definition 1.0.13 (Fixed Point)

A fixed point is invariant under a transformation such as a permutation.

2. Groups

2.0.1. Definition

Definition 2.0.1 (Group)

A *group* is a set G with a binary operation $* : G \times G \rightarrow G$ such that for all $a, b, c \in G$, it satisfies

- (i) (Closure) $a * b \in G$
- (ii) (Associativity) $(a * b) * c = a * (b * c)$
- (iii) (Identity) $\exists e \in G : e * a = a * e = a$
- (iv) (Inverse) $\exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$

Additionally, a group is said to be *abelian* or *commutative* if

- (v) (Commutative) $a * b = b * a$

💡 Tip

The proof of (iii) and (iv) can be shortened if (v) is proven.

📖 Note

Defying conventional wisdom, *abelian* is usually not capitalized.

Definition 2.0.2 (Subgroup)

Let $(G, *)$ be a group. A subset $H \subseteq G$ is a *subgroup* if it is itself a group under the operation inherited from G .

A common test is: H is non-empty and for all $a, b \in H$, the element $a * b^{-1}$ also lies in H .

2.1. PERMUTATION GROUPS

2.1.1. Symmetric Groups

Definition 2.1.1 (Symmetric Group)

The *symmetric group* of a set X with n elements is

$$S_n = \{\sigma : X \rightarrow X \mid \sigma \text{ is bijective}\}$$

and has $n!$ elements.

Definition 2.1.2 (Symmetric Group, Permutations)

The *symmetric group* S_n is the group of all permutations on n elements under the operation of composition.

2.1.2. Alternating Groups

Definition 2.1.3 (Alternating Group)

The *alternating group*

$$A_n = \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\}$$

is the subgroup of S_n consisting of all even permutations.

It has $n!/2$ elements for $n \geq 2$.

Theorem 2.1.1

The sign map $\text{sgn} : S_n \rightarrow \{1, -1\}$ is a group homomorphism.

The kernel of this homomorphism is the alternating group

$$\ker(\text{sgn}) = A_n.$$

3. Fields

Definition 3.1 (Field)

A *field* is a set F with two binary operations $+$ and $\cdot$ such that

- (i) $(F, +)$ is an abelian group with identity 0 .
- (ii) $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1 .
- (iii) Multiplication distributes over addition:

$$\text{for all } a, b, c \in F, \quad a \cdot (b + c) = a \cdot b + a \cdot c.$$

Examples. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{R}(x), \mathbb{Q}(i), \mathbb{F}_p$ if p is prime.

Definition 3.2 (Vector Space)

A vector space over a field F is a set V with two operations:

- Vector addition $+: V \times V \rightarrow V$
- Scalar multiplication $\cdot : F \times V \rightarrow V$

These must satisfy

- (i) (Closure under addition) $\mathbf{u} + \mathbf{v} \in V$
- (ii) (Distributivity over vectors) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- (iii) (Associativity) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- (iv) (Identity) $\mathbf{u} + \mathbf{0} = \mathbf{u}$

$$(v) \text{ (Inverse) } \mathbf{u} + (-\mathbf{u}) = \mathbf{0}$$

$$(vi) \text{ (Commutativity) } \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$(vii) \text{ (Closure under scalar multiplication) } a\mathbf{u} \in V$$

$$(viii) \text{ (Distributivity over scalars) } (a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$$

$$(ix) \text{ (Associativity) } a(b\mathbf{u}) = (ab)\mathbf{u}$$

$$(x) \text{ (Identity) } 1\mathbf{u} = \mathbf{u}$$

Note that requiring closure is often omitted.